

Might there be a rigorous proof that all analogies are dubious? Obviously not, because, as we just saw, everyone depends, without thinking, on a dense avalanche of mini-analogies between everyday things, and these mini-analogies follow on the heels of one another all day long, day in, day out — and seldom do such mundane analogies mislead anyone. Indeed, if they did, we would not be here to tell the tale.

Giant Electronic Dunces

How can computers be so terribly stupid, despite being so blindingly fast and having such huge and infallible memories? Contrariwise, how can human beings be so insightful despite being so limited in speed and having such small and fallible memories? Though perhaps hackneyed, these are reasonable and important questions, focusing as they do on the nearly paradoxical quality of human thought.

Indeed, the human mind, next to a computer, appears fraught with defects of every sort, coming off as hopelessly inferior along most dimensions of comparison. For instance, in carrying out pure reasoning tasks, well-polished computer algorithms reach logically valid conclusions virtually instantly, while people tend to fail most of the time. Much the same can be said about large amounts of knowledge. Where people's minds are saturated after only a few pieces of information are presented, a computer can take into account a virtually unlimited amount of information. And of course human memory is notoriously unreliable; whereas computers never forget and never distort, those are activities at which we human beings excel, for better or for worse. Three days, three weeks, three months, or three years after we've seen a movie or read a book, what details of it remain accessible in our minds? And how distorted are they? We might also mention the speed at which processing takes place in computers as opposed to human brains. What might take us minutes, hours, or far longer can be done by a computer in an infinitesimal eyeblink. Just consider simple arithmetical calculations such as " $3 + 5$ " (a bit under a second for a person), or " $27 + 92$ " (perhaps five or ten seconds), or " 27×92 ", a calculation that most people could not carry out in their heads. Counting the number of words in a selected passage of text and correcting a multiple-choice exam are activities that we humans can carry out, but only with pathetic slowness compared to computers.

Overall, the comparison is extremely lopsided in favor of computers, for, as we just noted, computers carry out flawless reasoning and calculation way beyond human reach, handle unimaginably larger amounts of information than people can handle, do not forget things over short or long time scales, do not distort what they memorize, and carry out their processing at speeds incomparably greater than that of the human mind. In terms of rationality, size, reliability, and speed, the machines we have designed and built beat us hands down. If we then add to the human side of the ledger our easily distractable attention, the fatigue that often seriously interferes with our capacities, and the imprecision of our sensory organs, we are left straggling in the dust. If one were to draw up a table of numerical specifications, as is standardly done in comparing one computer with another, *Homo sapiens sapiens* would wind up in the recycling bin.

Given all this, how can we explain the fact that, in terms of serious thought, machines lag woefully behind us? Why is machine translation so often inept and awkward? Why are robots so primitive? Why is computer vision restricted to the simplest kinds of tasks? Why is it that today's search engines can instantly search billions of Web sites for passages containing the phrase "in good faith", yet are incapable of spotting Web sites in which the *idea* of good faith (as opposed to the string of alphanumeric characters) is the central theme?

Readers will of course have anticipated the answer — namely, that our advantage is intimately linked to categorization through analogy, a mental mechanism that lies at the very center of human thought but at the furthest fringes of most attempts to realize artificial cognition. It is only thanks to this mental mechanism that human thoughts, despite their slowness and vagueness, are generally reliable, relevant, and insight-giving, whereas computer "thoughts" (if the word even applies at all) are extremely fragile, brittle, and limited, despite their enormous rapidity and precision.

As soon as categorization enters the scene, the competition with computers takes on a new kind of lopsidedness — but this time greatly in favor of humans. The primordial importance of categorization through analogy in helping living organisms survive becomes obvious if one tries to imagine what it would be like to "perceive" the world in a manner entirely devoid of categories — something like how the world must appear to a newborn, for whom each new concept has to be acquired from scratch and with great difficulty. By contrast, seeing the new in terms of the old and familiar allows one to benefit, and at only a slight cognitive cost, from knowledge previously acquired. Thus, if there were two creatures, one of which (an adult human being) perceived the world using categorization through analogy while the other (a computer) had no such mechanism to help it out, their competition in understanding the world around them would be comparable to a race between a person and a robot to climb up to a high roof, with the human allowed to use a preexistent staircase but with the robot required to construct its own staircase from scratch.

Analogy Operating at All Levels

Categorization through analogy drives thinking at all levels, from the smallest to the largest. Consider a conversation in which several hierarchical linguistic levels are continually interacting. First of all, the choice of a specific word will of course determine the sounds that make it up; similarly, when one is typing at one's keyboard, each word chosen determines the letters composing it, so that they come along automatically rather than being chosen one by one. Analogously, words are often determined by larger structures of which they are but pieces. This happens most clearly whenever one uses a stock phrase (such as "so to speak" or "cut to the chase" or "down to the wire" or "when push comes to shove" or "as easy as stealing candy from a baby"), but it also often happens when no such expression is involved, because one is always working under the constraints of the syntactic and semantic patterns of the language one is speaking, as well as those of one's own habitual speech patterns.